

ECON 139: Labor Economics

Problem Set 3

Note on Grading: Homework is graded using check +, check or check -. If you answer all questions and generally show you have put thought and effort into the problem set, you will get a check + (1/1). If your answers are sometimes incomplete, or show a clear lack of effort in parts, you will receive a check (0.7/1). If your answers are often incomplete and show little effort overall, you will get a check - (0/1). The total number of points for each problem set in the course is 1 point and each problem set is worth 5 percent of the grade. If you turn this in within one day late, the highest score you can get is a check (0.7). No credit is given for assignments that are turned in more than one day late.

Note on AI Use: As discussed in the syllabus, you can use AI tools to help with this problem set. For the first part, however, I highly recommend you complete this on your own without any AI assistance. This type of problem may very well appear on a midterm or final. If you cannot solve it yourself, you will likely struggle on the exam.

For the empirical part, I recommend using AI tools. Not everything in the empirical question comes directly from the lecture. You will need to figure out some parts on your own (as you will often need to do in the future!). AI tools will be useful for this.

1 Part 1: Theoretical Exercise

1. Imagine worker A faces a wage rate of \$15 per hour and can work up to 16 hours per day. Draw the budget constraint for this worker assuming they have no non-labor income. Label the axes and the intercepts.
2. Draw an indifference curve for this worker that is consistent with them choosing to work 8 hours per day.
3. Now imagine the wage rate for overtime work is equal to \$20 per hour for any hours worked beyond 8 hours per day. Draw the new budget constraint for this worker. Label the axes and intercepts.
4. Given the new budget constraint, will the worker choose to work more than 8 hours, less than 8 hours, or exactly 8 hours? Explain your reasoning.
5. For this question, you can remove the overtime as a consideration. Now, instead, imagine the government combines a Universal Basic Income (UBI) of \$60 per day to all workers. However, some are concerned that this UBI will reduce their incentive to work and want to combine this with a work requirement, where you must work at least 4 hours per day to receive the

UBI. Draw the new budget constraint for the worker with the original wage of \$15 per hour and the UBI of \$60 per day that is only available to those who work at least 4 hours. Label the axes and intercepts.

6. In response to this particular design of UBI, is it possible some individuals lower their total hours supplied while others increase it? Answer using indifference curve diagrams. Explain your reasoning. Remember, it is completely possible that individuals have different preferences, so you can assume there are likely individuals who choose the entire range of different hours options.

2 Part 2: Empirical Exercise

In this empirical exercise, we are going to use a subset of the American Community Survey (ACS) data to analyze hours across education groups, age, and sex. The data is available on Canvas under the Problem Set 3.

The data contains 50,000 observations from the 2023 ACS (the data contains the same variables that are in the occupation presentation version of the data, so you will likely already be familiar with the dataset). The goal of this exercise is to produce figures that are informative on how hours worked per week varies across education groups, age, and sex. For all figures, you should make sure to label the axes and provide a descriptive title. Like the occupation presentation figures, these should be well-formatted and easy to read. If you are using a coding-based program, you just need to submit your code. You don't need to submit the figures. If you are using Excel, the spreadsheet should include the figures. They can be in separate sheets if that is easier, but they should be easily accessible.

1. The variable "educ" contains codes for different levels of education. Using this variable, create a new variable that categorizes individuals into three groups: "Less than High School", "High School Graduate", and "College". You will notice in the data that there are individuals with 1 year of college, 2 years of college, etc. For this exercise, anyone with any college experience should be categorized as "College". Remember, if you want to understand what numeric codes correspond to, you can always google "IPUMS ACS varname". So for this variable, you can google "IPUMS ACS educ" to find the codebook.
2. Create a bar graph that shows the average hours worked per week (variable "uhrswork") for each of the three education groups you created in part (a). Make sure to label the axes and provide a descriptive title.
3. Next, we want to see how hours worked varies by age and sex. First, restrict to individuals between 18 and 75. Create a line graph that shows the average of "uhrswork" by age, with separate lines for male and female individuals. Therefore, the x-axis should be age, the y-axis should be average hours, and there should be two separate lines on the figure. Make sure to label the axes and provide a descriptive title. Again, you don't need to write anything down, but think about whether you find anything surprising or interesting in this figure.